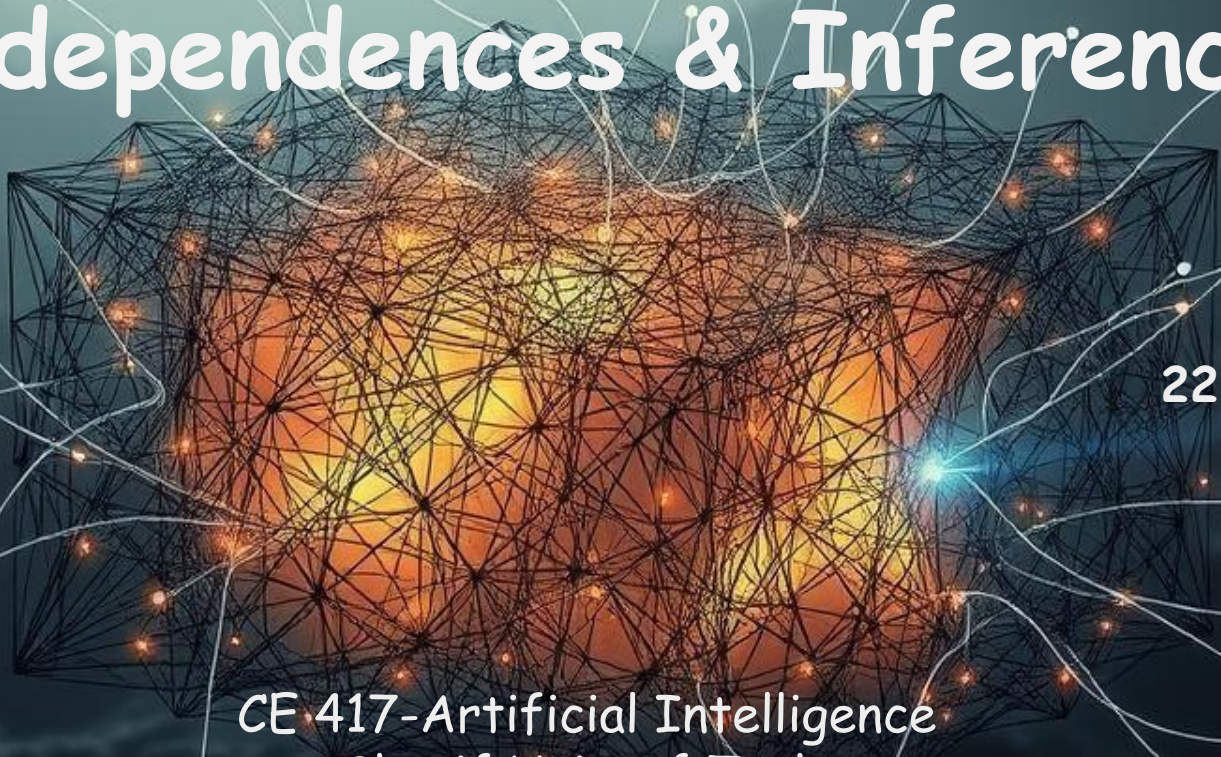


Bayes' Nets: Independences & Inference

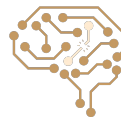


Arash
Marioriyad

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1405

CE 417-Artificial Intelligence
Sharif Univ. of Tech.

Some of Slides have been adopted from Berkeley CS188 &
Sharif UT CE417



Joint Distributions

- A joint distribution over a set of random variables: $X_1, X_2, \dots, X_n$ specifies a real number for each assignment (or outcome):

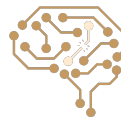
$$P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n)$$

$$P(x_1, x_2, \dots, x_n)$$

- Must obey:

$$P(x_1, x_2, \dots, x_n) \geq 0$$

$$\sum_{(x_1, x_2, \dots, x_n)} P(x_1, x_2, \dots, x_n) = 1$$

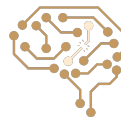


The Chain Rule

- Any joint distribution can be considered as an incremental product of conditional distributions!

$$P(x_1, x_2, x_3) = P(x_1)P(x_2|x_1)P(x_3|x_1, x_2)$$

$$P(x_1, x_2, \dots x_n) = \prod_i P(x_i|x_1 \dots x_{i-1})$$

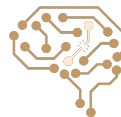


Independence

- Two variables are independent if:

$$\forall x, y : P(x, y) = P(x)P(y)$$

- This says that their joint distribution factors into a product two simpler distributions
- Another form: $\forall x, y : P(x|y) = P(x)$
- We write: $X \perp\!\!\!\perp Y$



Conditional Independence

- X is conditionally independent of Y given Z

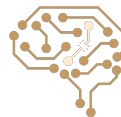
$$X \perp\!\!\!\perp Y | Z$$

- If and only if:

$$\forall x, y, z : P(x, y | z) = P(x | z) P(y | z)$$

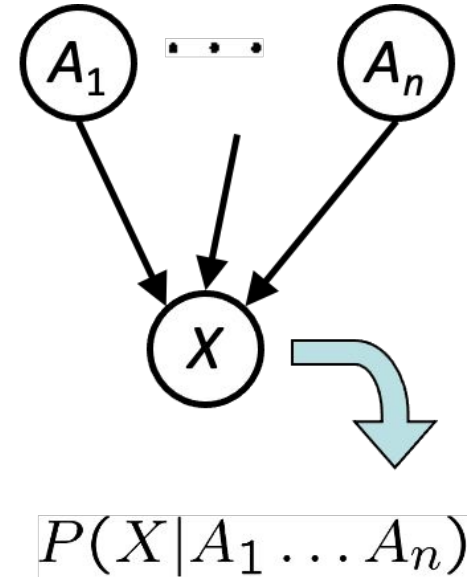
- Or, equivalently, if and only if:

$$\forall x, y, z : P(x | z, y) = P(x | z)$$



Bayes' Net

- A directed acyclic graph (DAG) with a set of nodes one per variable
- Each link from X to Y shows a "direct influence" of X on Y
- For each node, a conditional probability distribution $P(X_i | \text{Parents}(X_i))$ by CPT (conditional probability table)



A Bayes net = Topology (graph) + Local Conditional Probabilities



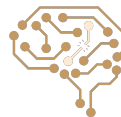
Probabilities in BNs

- Bayes' nets implicitly encode joint distributions
- Chain Rule:

$$P(x_1, x_2, \dots x_n) = \prod_i P(x_i | x_1 \dots x_{i-1})$$

- Bayes Net Version:

$$P(x_1, x_2, \dots x_n) = \prod_{i=1}^n P(x_i | \text{parents}(X_i))$$



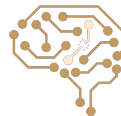
Two Equivalent Assumptions of Bayes Nets

- Factorization Property:

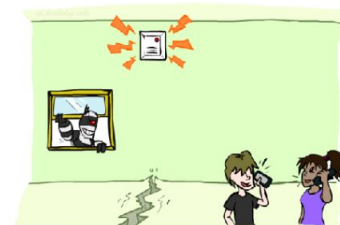
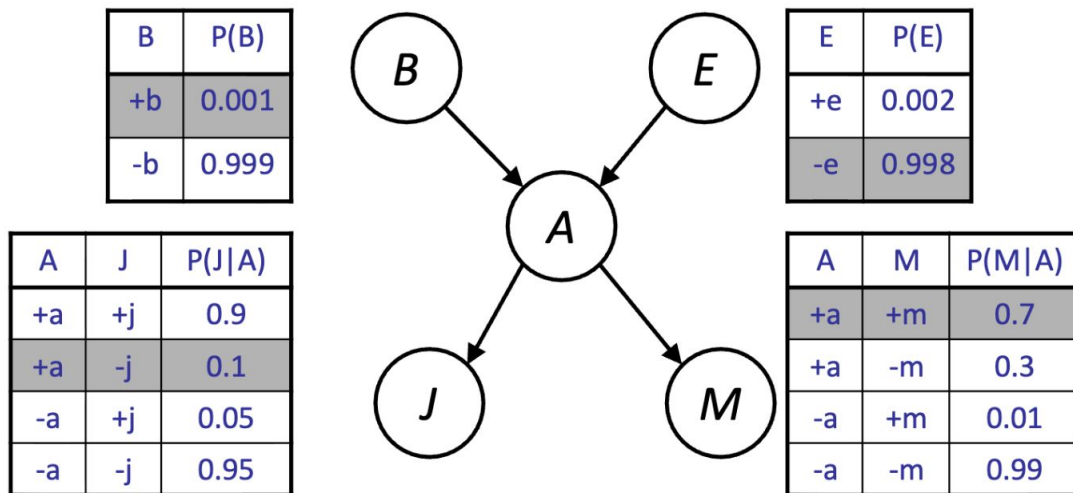
$$P(x_1, x_2, \dots x_n) = \prod_{i=1}^n P(x_i | \text{parents}(X_i))$$

- Local Markov Property:

$$X_i \perp \text{NonDescendants}(X_i) \mid \text{Parents}(X_i)$$

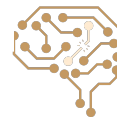


Example: Alarm Network (cont.)



B	E	A	P(A B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

$$\begin{aligned}
 P(+b, -e, +a, -j, +m) &= \\
 P(+b)P(-e)P(+a|+b, -e)P(-j|+a)P(+m|+a) &= \\
 0.001 \times 0.998 \times 0.94 \times 0.1 \times 0.7 &=
 \end{aligned}$$

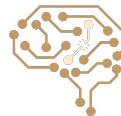
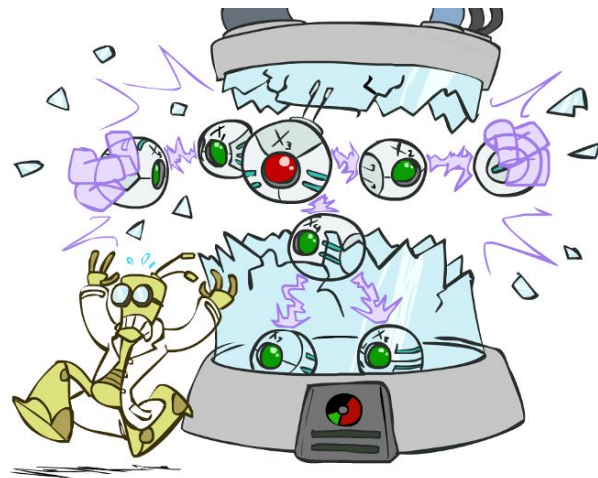


Today's Content: Two Types of Queries

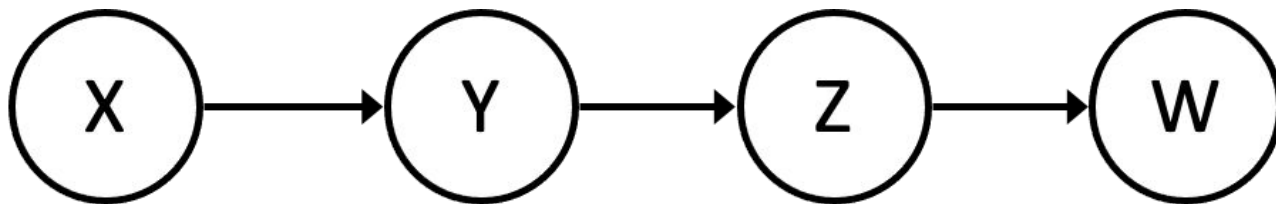
- Asking about Conditional Independencies:
 - Is X conditionally independent of Y give {Z}?

- Probabilistic Inference:

$$P(Q|E_1 = e_1, \dots, E_k = e_k) ?$$



Example: Conditional Independence in BNs



- Conditional independence assumptions directly from the topology?

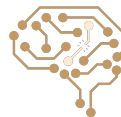
$$Z \perp\!\!\!\perp X|Y$$

$$W \perp\!\!\!\perp Y|Z$$

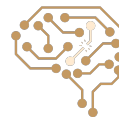
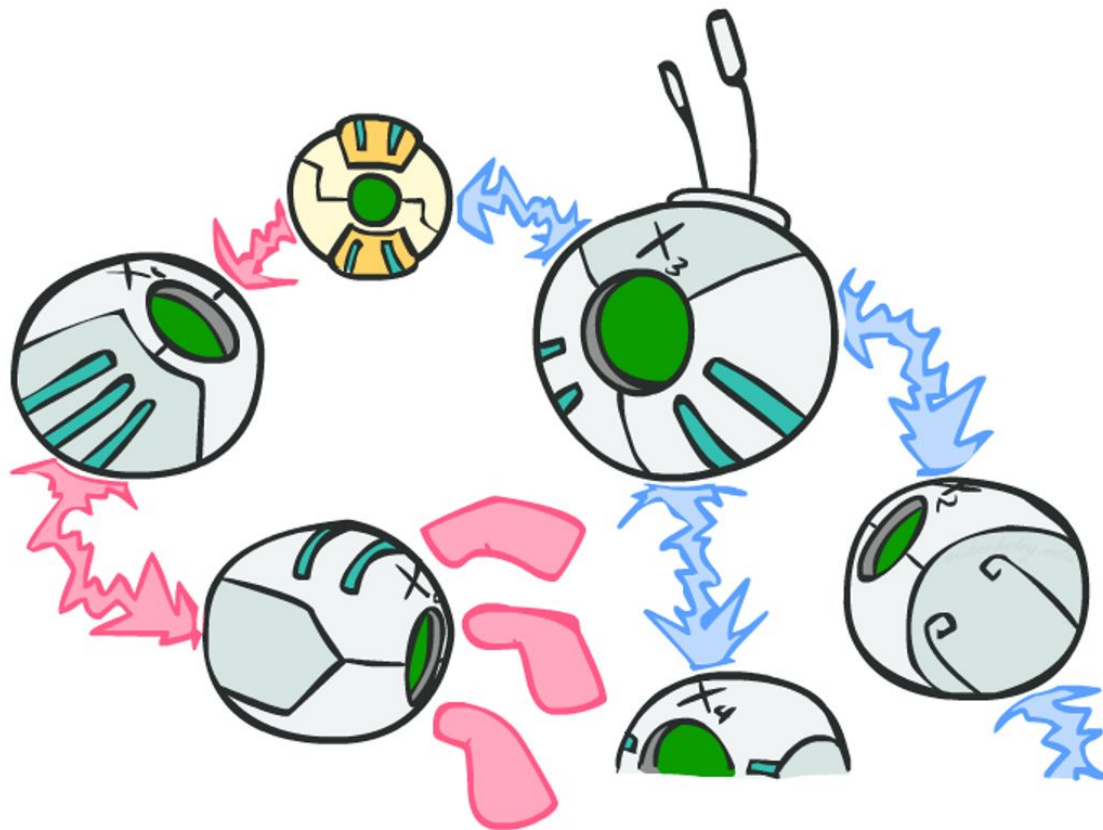
$$W \perp\!\!\!\perp X|Z$$

- Additional implied conditional independencies?

$$W \perp\!\!\!\perp X|Y$$



D-separation Algorithm



Types of Relations in a Triples (X, Y, Z)

- Causal Chain
- Common Cause
- Common Effect



Causal Chains



X: Low pressure

Y: Rain

Z: Traffic

$$P(x, y, z) = P(x)P(y|x)P(z|y)$$

Guaranteed X independent of Z? **No!**

One example set of CPTs for which X is not independent of Z is sufficient to show this independence is not guaranteed.

Example: Low pressure causes rain causes traffic, high pressure causes no rain causes no traffic

$$P(+y | +x) = 1, P(-y | -x) = 1,$$

$$P(+z | +y) = 1, P(-z | -y) = 1$$



Causal Chains



X: Low pressure

Y: Rain

Z: Traffic

$$P(x, y, z) = P(x)P(y|x)P(z|y)$$

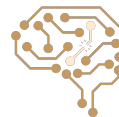
Guaranteed X independent of Z given Y? **YES!**

$$P(z|x, y) = \frac{P(x, y, z)}{P(x, y)}$$

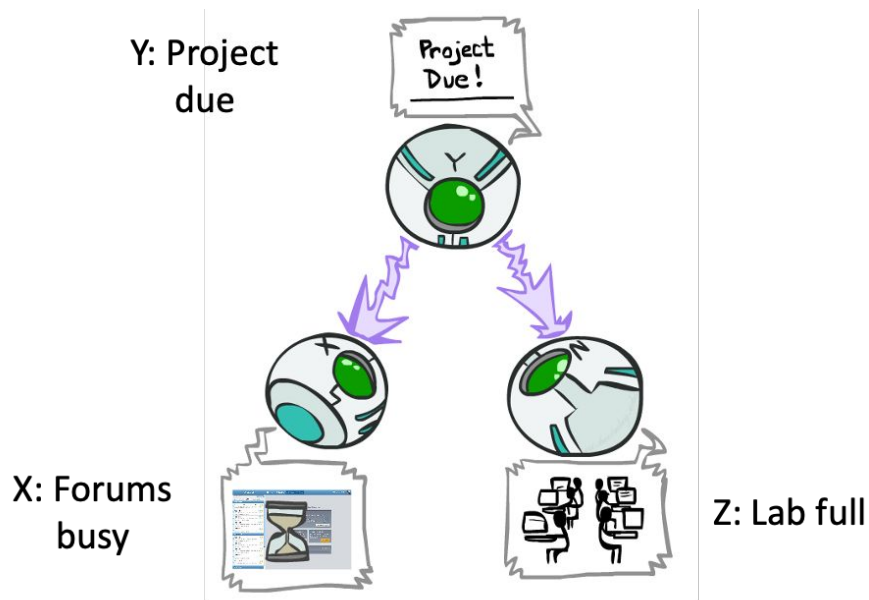
$$= \frac{P(x)P(y|x)P(z|y)}{P(x)P(y|x)}$$

$$= P(z|y)$$

Evidence along the chain “blocks” the influence



Common Cause



$$P(x, y, z) = P(y)P(x|y)P(z|y)$$

Guaranteed X independent of Z? **No!**

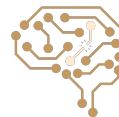
One example set of CPTs for which X is not independent of Z is sufficient to show this independence is not guaranteed.

Example:

Project due causes both forums busy and lab full

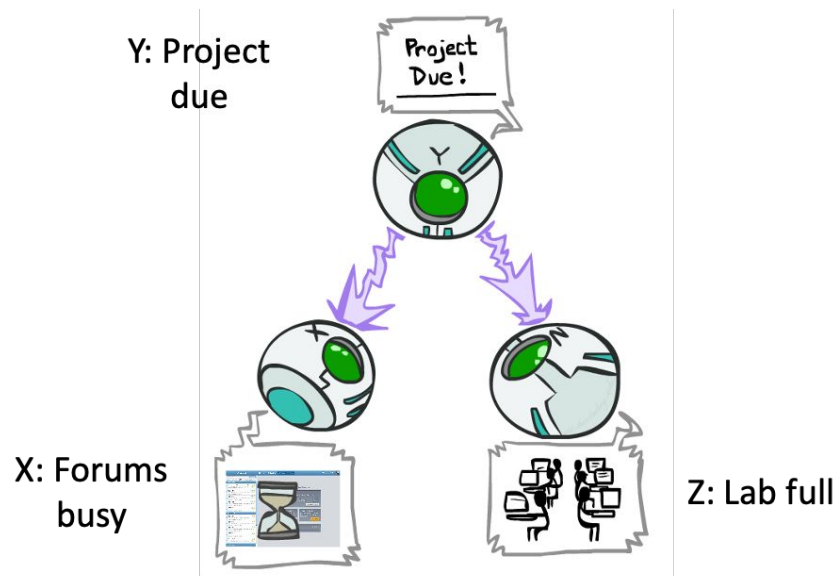
$$P(+x \mid +y) = 1, P(-x \mid -y) = 1,$$

$$P(+z \mid +y) = 1, P(-z \mid -y) = 1$$



Common Cause

Guaranteed X independent of Z given Y? **YES!**



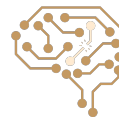
$$P(x, y, z) = P(y)P(x|y)P(z|y)$$

$$P(z|x, y) = \frac{P(x, y, z)}{P(x, y)}$$

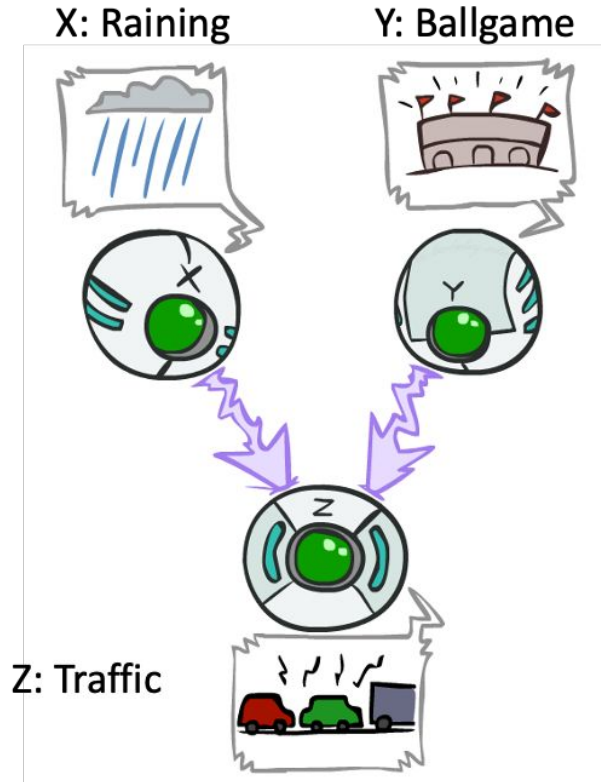
$$= \frac{P(y)P(x|y)P(z|y)}{P(y)P(x|y)}$$

$$= P(z|y)$$

Observing the cause blocks influence between effects.



Common Effect



Are X and Y Independent? **YES**

The ballgame and the rain cause traffic, but they are not correlated

Are X and Y Independent given Z? **NO**

Seeing traffic puts the rain and the ballgame in competition as explanation.

Observing an effect activates influence between possible causes.



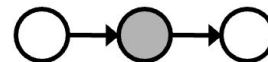
Recap of Triples

Causal Chain:

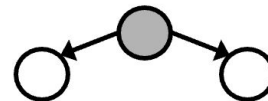
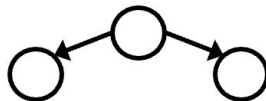
Active Triples



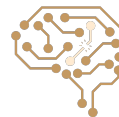
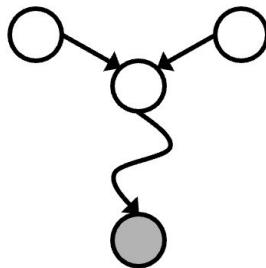
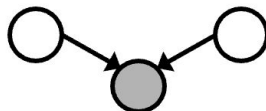
Inactive Triples



Common Cause:

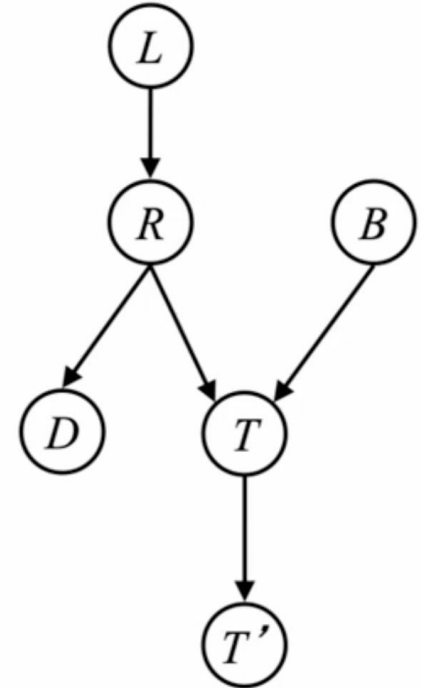


Common Effect ("v-structure")



General Query

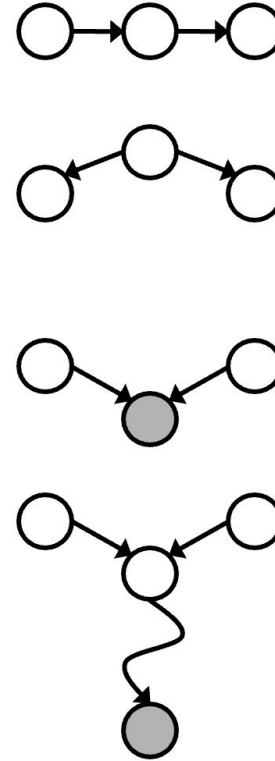
- In a given BN, are two variables independent given some evidences?
- Example: Is L conditionally independent of T given R and B as evidences?
- Any complex example can be broken into repetitions of the three canonical cases



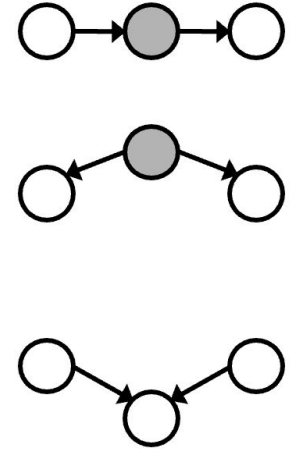
Active and Inactive Paths

- A **path** consists of a number of triples
- Example: for a path **A–B–C–D–E**, the triples are:
 - A–B–C
 - B–C–D
 - C–D–E
- A path is active if each triple is active

Active Triples



Inactive Triples



D-Separation: Answering to General Queries

Question: Are X and Y conditionally independent given evidence $\{Z\}$?

- Consider all (undirected) paths between X and Y
- If no any active path between X and $Y \rightarrow X$ and Y “d-separated” by $Z \rightarrow$ Conditionally Independent
- If at least one active path, then independence is not guaranteed!

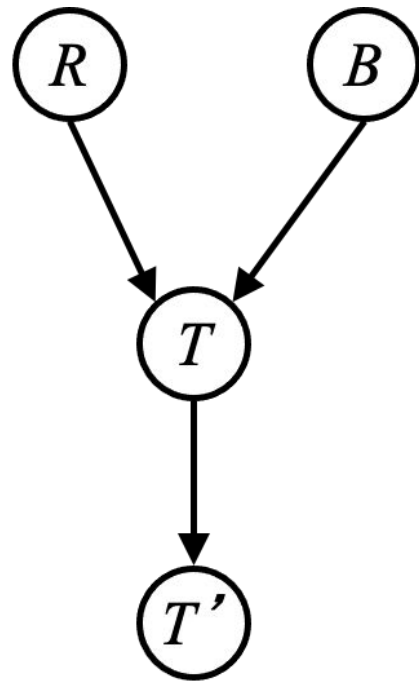


Quiz 1

$$R \perp\!\!\!\perp B$$

$$R \perp\!\!\!\perp B | T$$

$$R \perp\!\!\!\perp B | T'$$



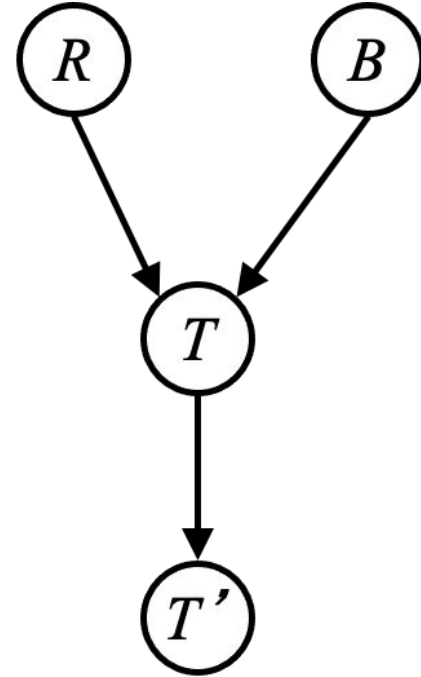
Quiz 1 - Answer

$$R \perp\!\!\!\perp B$$

Yes

$$R \perp\!\!\!\perp B | T$$

$$R \perp\!\!\!\perp B | T'$$



Quiz 2

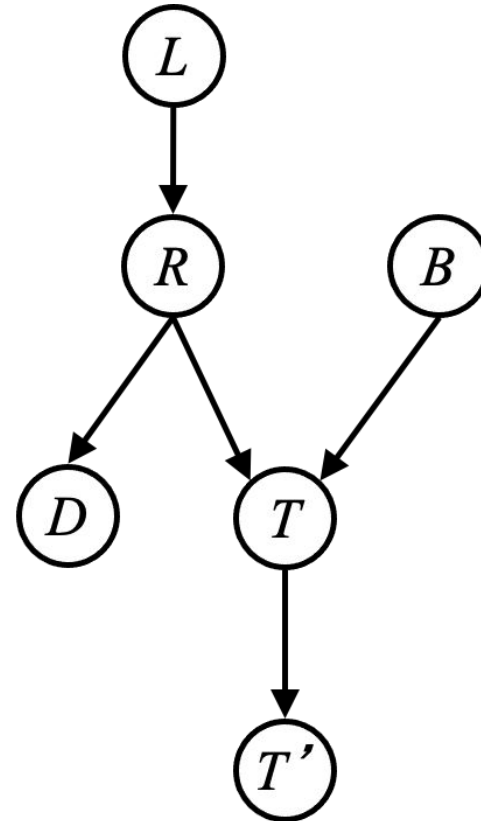
$$L \perp\!\!\!\perp T' | T$$

$$L \perp\!\!\!\perp B$$

$$L \perp\!\!\!\perp B | T$$

$$L \perp\!\!\!\perp B | T'$$

$$L \perp\!\!\!\perp B | T, R$$



Quiz 2 - Answer

$$L \perp\!\!\!\perp T' | T$$

Yes

$$L \perp\!\!\!\perp B$$

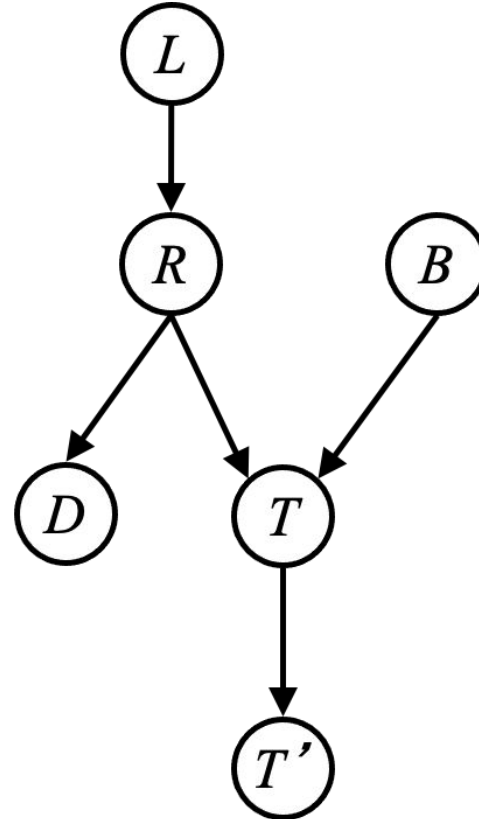
Yes

$$L \perp\!\!\!\perp B | T$$

$$L \perp\!\!\!\perp B | T'$$

$$L \perp\!\!\!\perp B | T, R$$

Yes



Quiz 3

Variables:

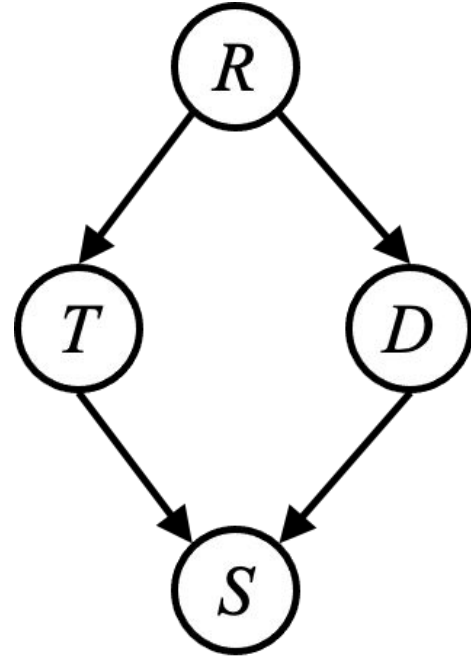
- R: Raining
- T: Traffic
- D: Roof drips
- S: I'm sad

Queries:

$$T \perp\!\!\!\perp D$$

$$T \perp\!\!\!\perp D | R$$

$$T \perp\!\!\!\perp D | R, S$$



Quiz 3 - Answer

Variables:

- R: Raining
- T: Traffic
- D: Roof drips
- S: I'm sad

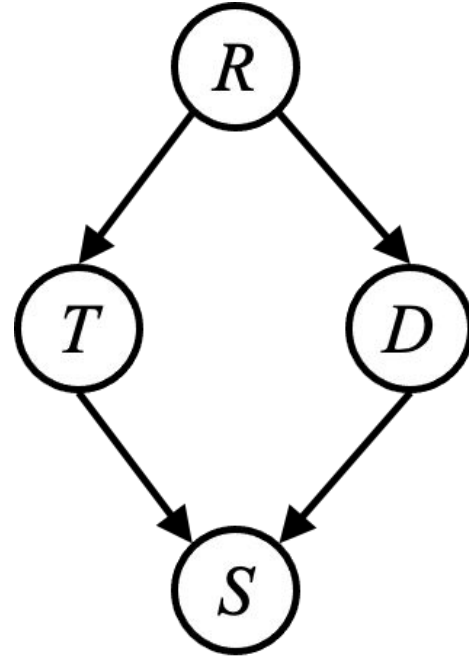
Queries:

$$T \perp\!\!\!\perp D$$

$$T \perp\!\!\!\perp D | R$$

Yes

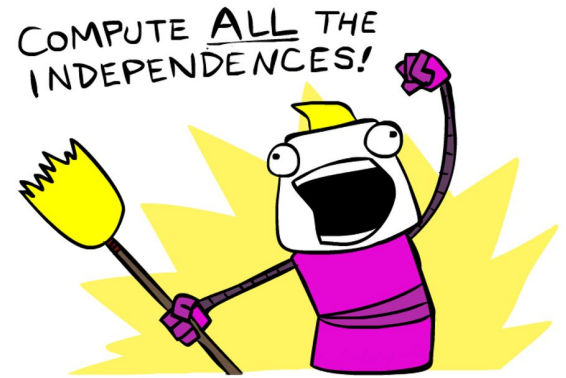
$$T \perp\!\!\!\perp D | R, S$$



Structure Implications

Given a Bayes net structure, can run d-separation algorithm to build a complete list of conditional independencies that are necessarily true of the form:

$$X_i \perp\!\!\!\perp X_j | \{X_{k_1}, \dots, X_{k_n}\}$$



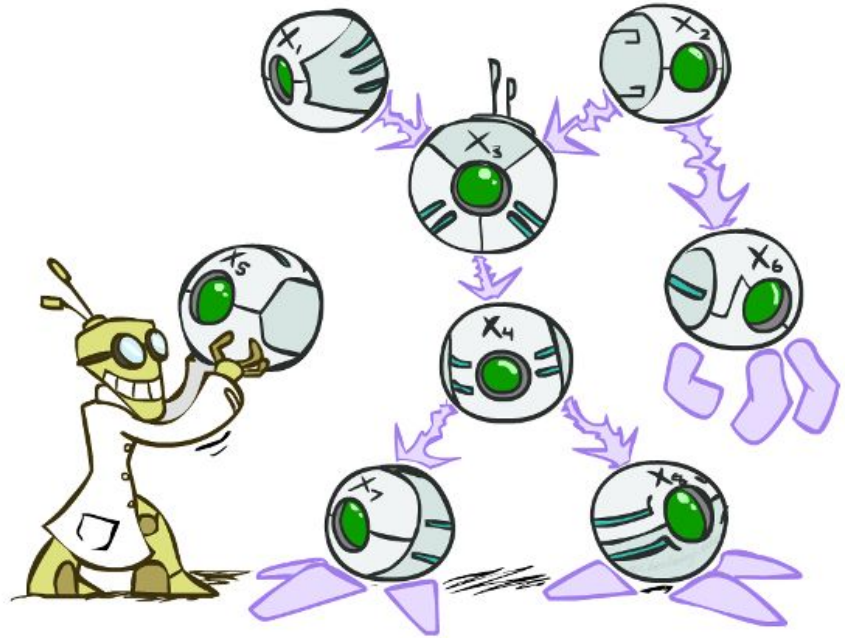
Bayes' Nets

- ✓ Representation
- ✓ Conditional Independencies

Probabilistic Inference

Exact

Approximate



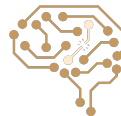
Inference

- Inference: calculating some useful quantity from a joint probability distribution

$$P(Q|E_1 = e_1, \dots, E_k = e_k)$$

- General case:

- Evidence variables: $E_1 \dots E_k = e_1 \dots e_k$
 - Query* variable: Q
 - Hidden variables: $H_1 \dots H_r$
- } $X_1, X_2, \dots, X_n$
All variables



Inference by Enumeration

- Step 1: Select the entries consistent with the evidence

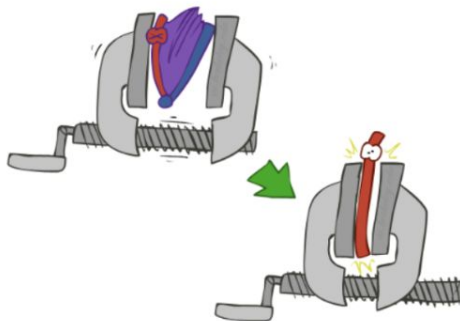


x	P(x)
-3	0.05
-1	0.25
0	0.07
1	0.2
5	0.01

2

0.15

- Step 2: Sum out H to get joint of Query and evidence



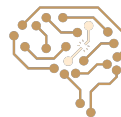
$$P(Q, e_1 \dots e_k) = \sum_{h_1 \dots h_r} P(Q, \underbrace{h_1 \dots h_r}_{X_1, X_2, \dots X_n}, e_1 \dots e_k)$$

- Step 3: Normalize

$$\times \frac{1}{Z}$$

$$Z = \sum_q P(Q, e_1 \dots e_k)$$

$$P(Q|e_1 \dots e_k) = \frac{1}{Z} P(Q, e_1 \dots e_k)$$



Inference by Enumeration in Bayes' Net

Given unlimited time, inference in BNs is easy

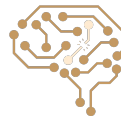
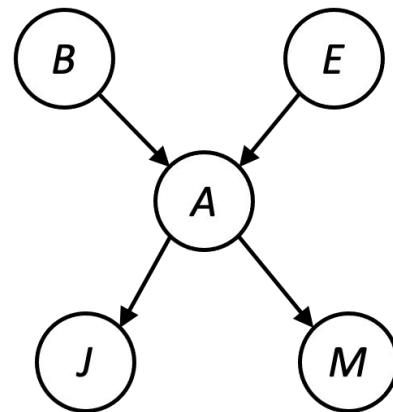
Reminder of inference by enumeration by example:

$$P(B \mid +j, +m) \propto_B P(B, +j, +m)$$

$$= \sum_{e,a} P(B, e, a, +j, +m)$$

$$= \sum_{e,a} P(B)P(e)P(a|B, e)P(+j|a)P(+m|a)$$

$$\begin{aligned} &= P(B)P(+e)P(+a|B, +e)P(+j|+a)P(+m|+a) + P(B)P(+e)P(-a|B, +e)P(+j|-a)P(+m|-a) \\ &\quad P(B)P(-e)P(+a|B, -e)P(+j|+a)P(+m|+a) + P(B)P(-e)P(-a|B, -e)P(+j|-a)P(+m|-a) \end{aligned}$$



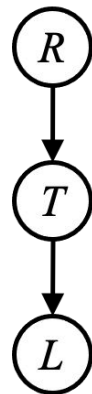
Example: Traffic Domain

Random Variables

R: Raining

T: Traffic

L: Late for class!



$$P(L) = ?$$

$$= \sum_{r,t} P(r, t, L)$$

$$= \sum_{r,t} P(r)P(t|r)P(L|t)$$

$$P(R)$$

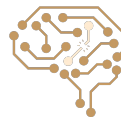
+r	0.1
-r	0.9

$$P(T|R)$$

+r	+t	0.8
+r	-t	0.2
-r	+t	0.1
-r	-t	0.9

$$P(L|T)$$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9



Inference by Enumeration: Procedural Outline

Track objects called **factors**

Initial factors are local CPTs (one per node)

$$P(R)$$

+r	0.1
-r	0.9

$$P(T|R)$$

+r	+t	0.8
+r	-t	0.2
-r	+t	0.1
-r	-t	0.9

$$P(L|T)$$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9

Any known values are selected

- E.g. if we know $L = +l$, the initial factors are

$$P(R)$$

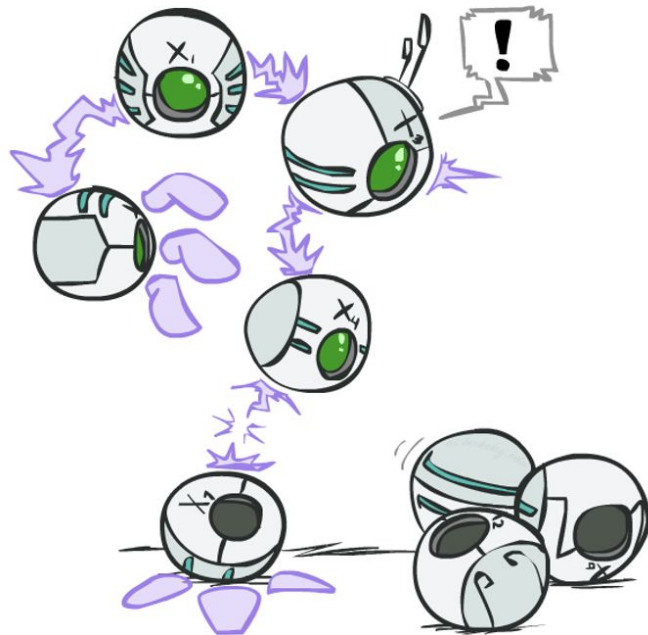
+r	0.1
-r	0.9

$$P(T|R)$$

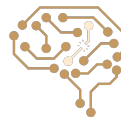
+r	+t	0.8
+r	-t	0.2
-r	+t	0.1
-r	-t	0.9

$$P(+l|T)$$

+t	+l	0.3
-t	+l	0.1



Procedure: Join all factors, then eliminate all hidden variables

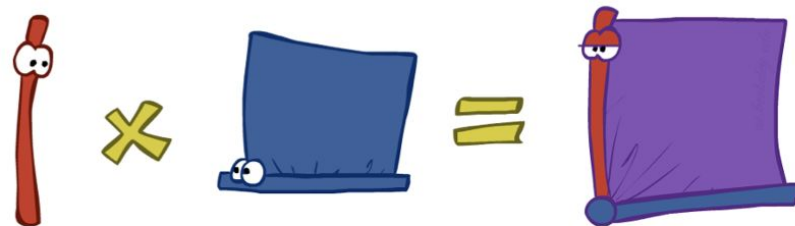


Operation 1: Join Factors

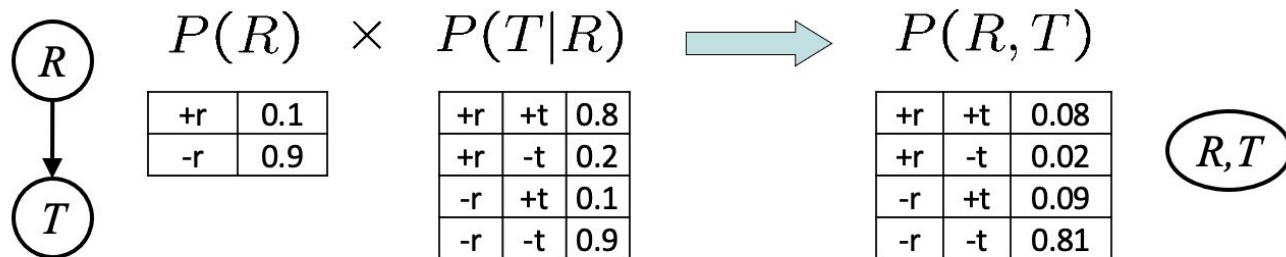
First basic operation: **joining factors**

Combining factors:

- **Just like a database join**
- Get all factors over the joining variable
- Build a new factor over the union of the variables involved



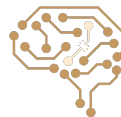
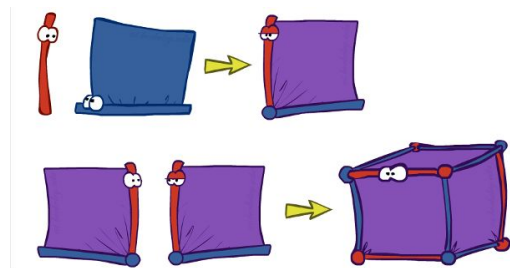
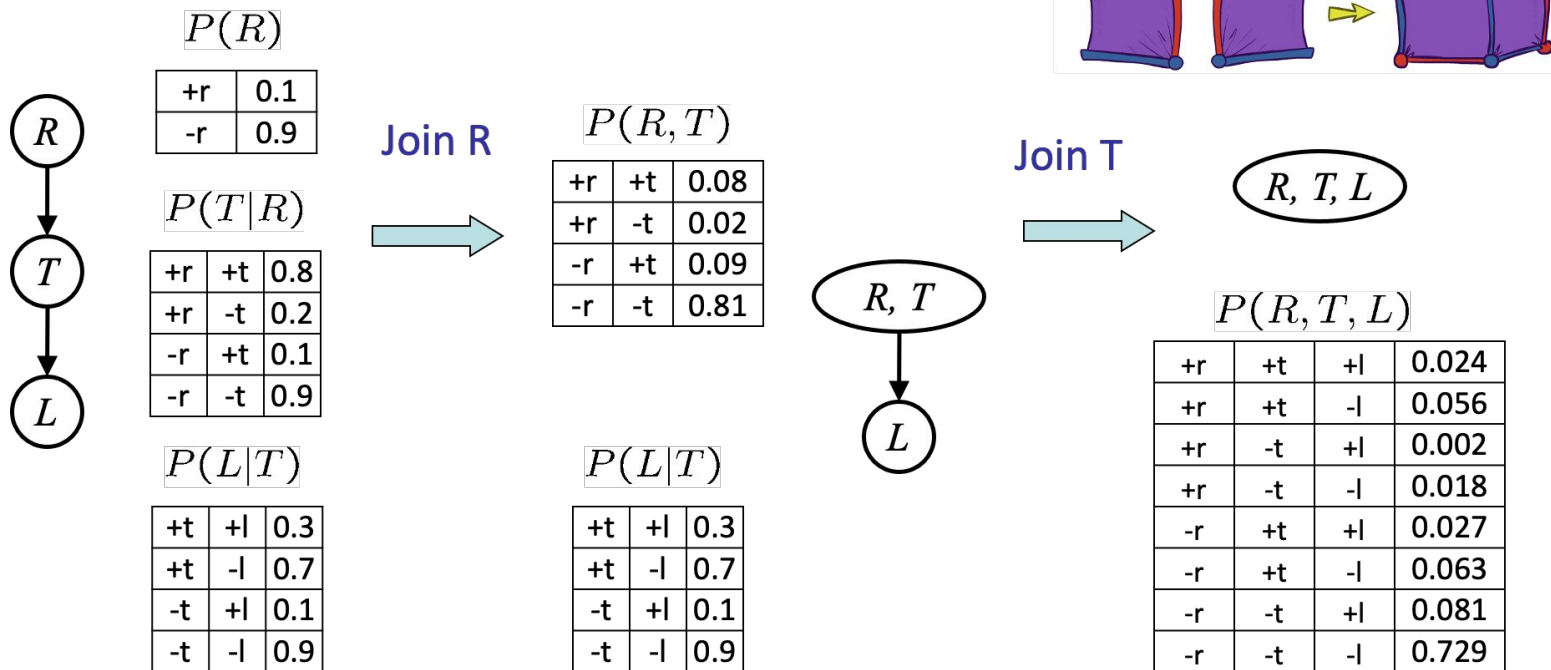
Example: Join on R



- Computation for each entry: pointwise products $\forall r, t : P(r, t) = P(r) \cdot P(t|r)$



Example: Multiple Joins



Operation 2: Eliminate (Marginalization)

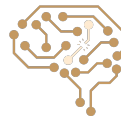
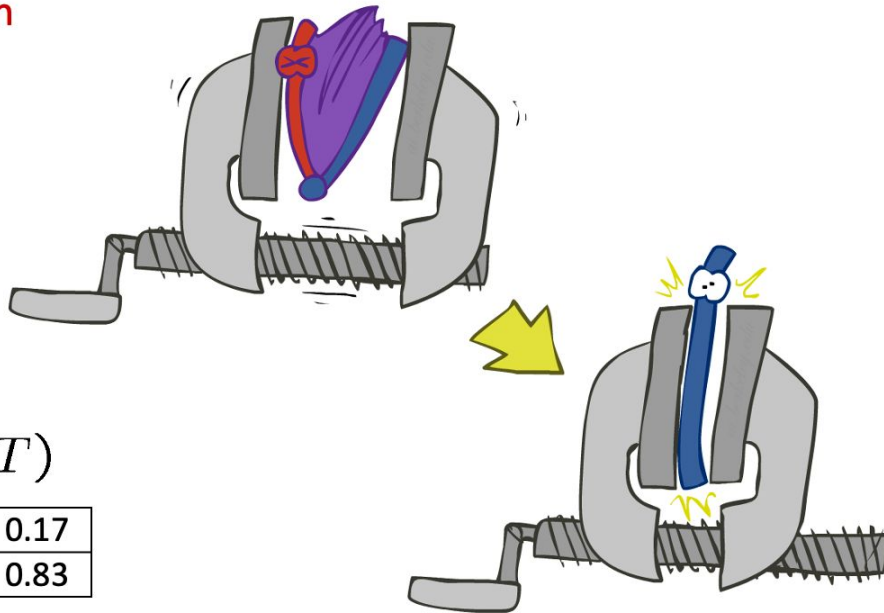
- Second basic operation: **marginalization**
- Take a factor and sum out a variable
 - Shrinks a factor to a smaller one
 - A **projection** operation
- Example:

$P(R, T)$		
+r	+t	0.08
+r	-t	0.02
-r	+t	0.09
-r	-t	0.81

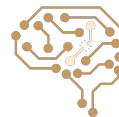
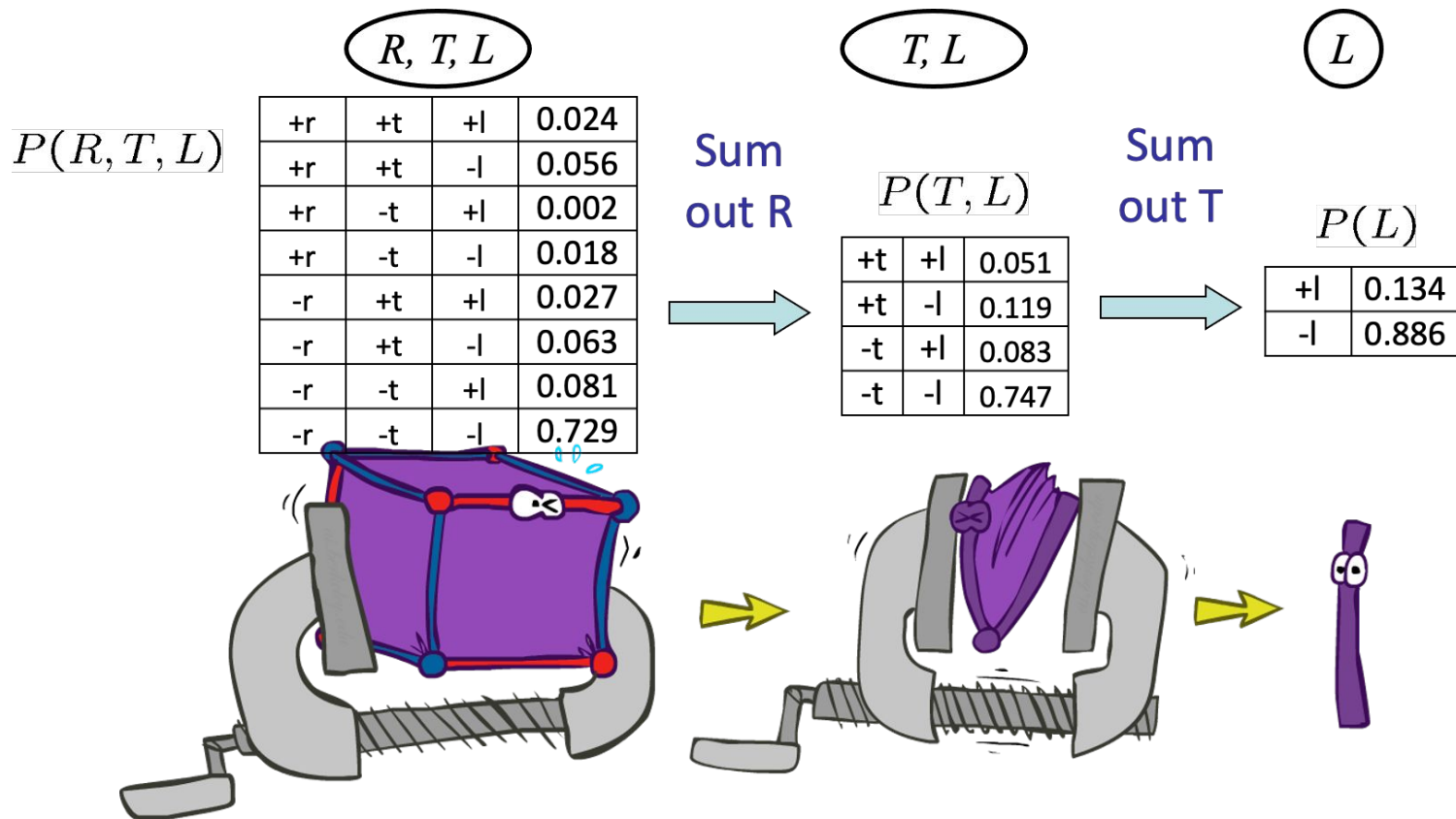
sum R



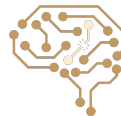
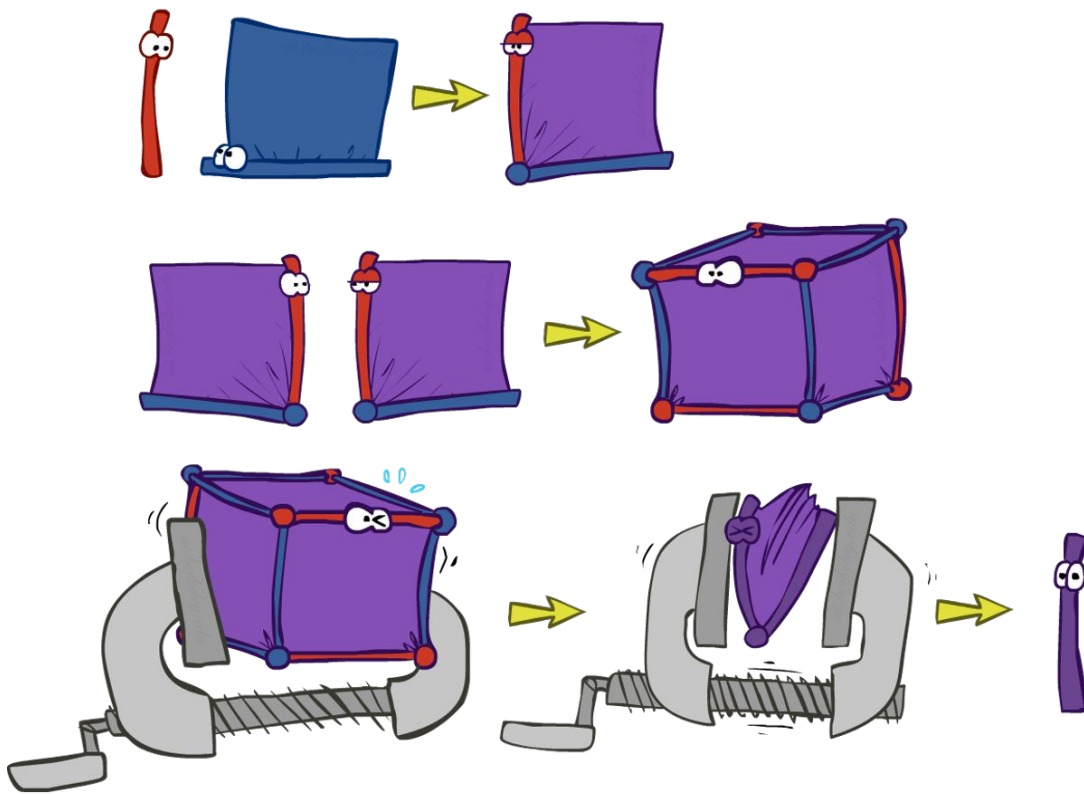
$P(T)$	
+t	0.17
-t	0.83



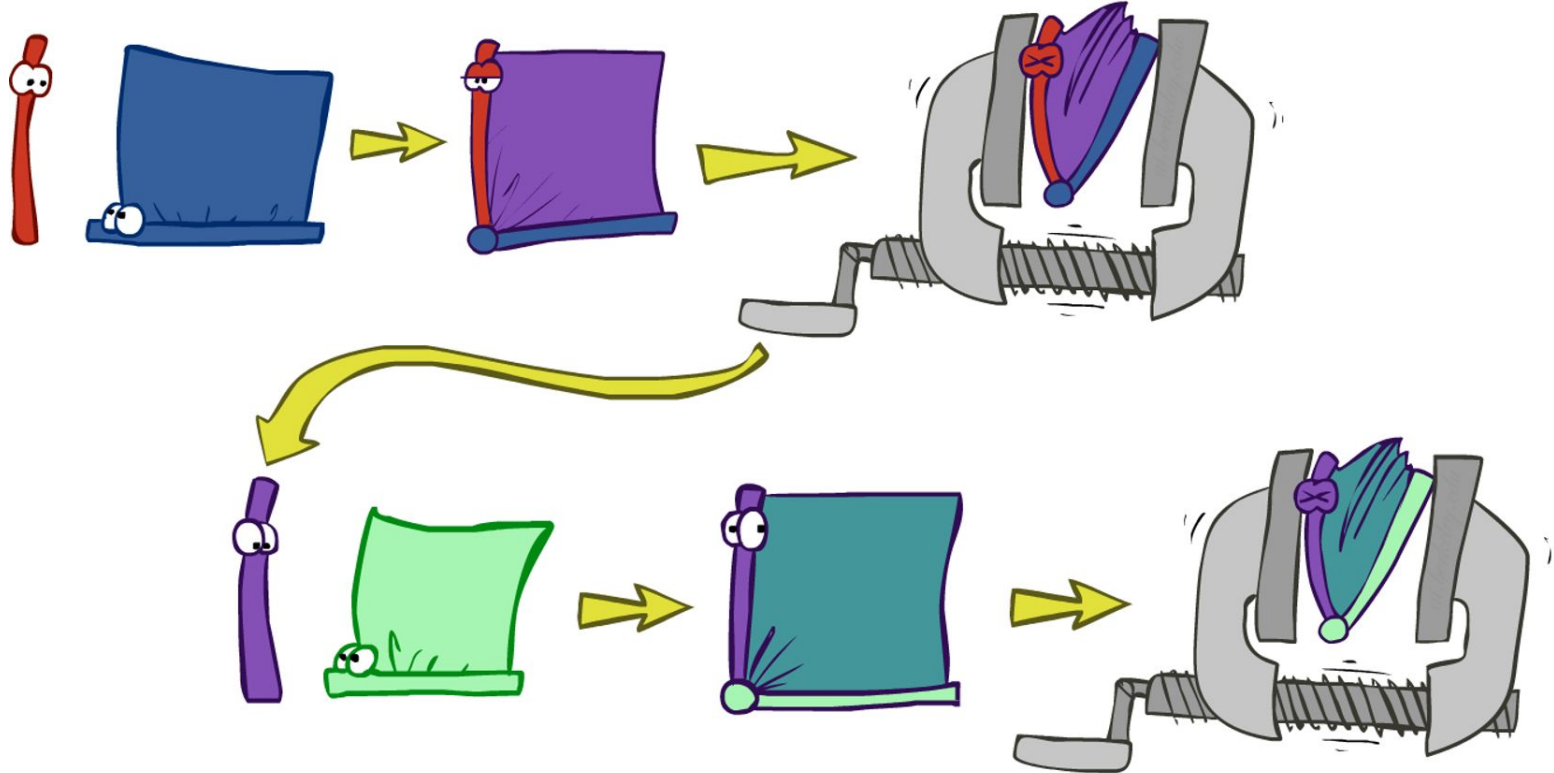
Multiple Elimination



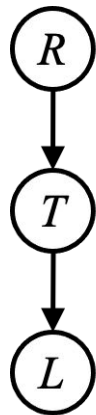
Inference by Enumeration = Multiple Join + Multiple Eliminate



Variable Elimination = Marginalizing Early



Traffic Domain



$$P(L) = ?$$

■ Inference by Enumeration

$$= \sum_t \sum_r P(L|t) \underbrace{P(r)P(t|r)}_{\text{Join on } r}$$

Join on t

Eliminate r

Eliminate t

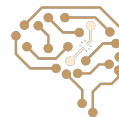
■ Variable Elimination

$$= \sum_t P(L|t) \sum_r \underbrace{P(r)P(t|r)}_{\text{Join on } r}$$

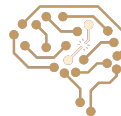
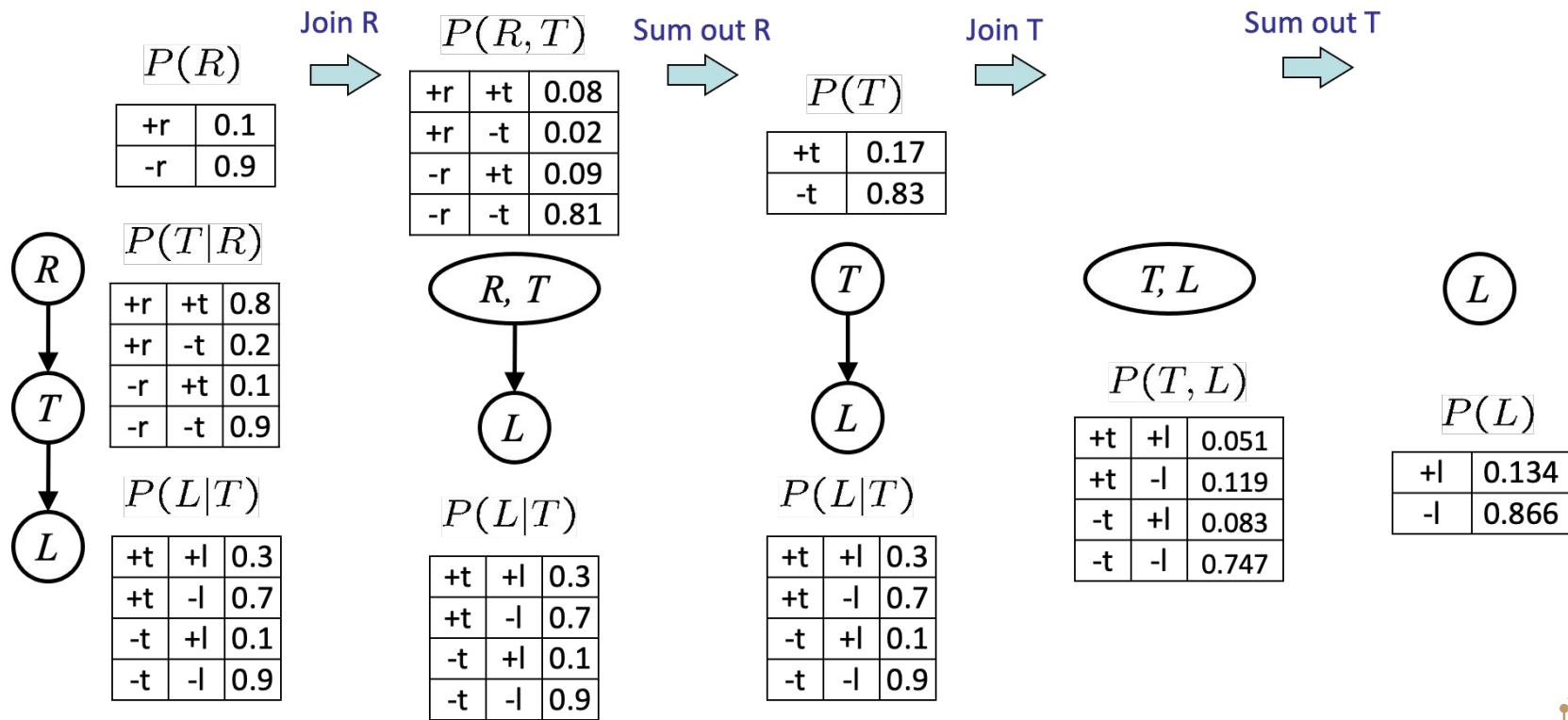
Eliminate r

Join on t

Eliminate t



Marginalizing Early! (aka VE)



Value Elimination: Procedural Outline

- If evidence, start with factors that select that evidence

- No evidence uses these initial factors:

$$P(R)$$

+r	0.1
-r	0.9

$$P(T|R)$$

+r	+t	0.8
+r	-t	0.2
-r	+t	0.1
-r	-t	0.9

$$P(L|T)$$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9

- Computing $P(L|+r)$ the initial factors become:

$$P(+r)$$

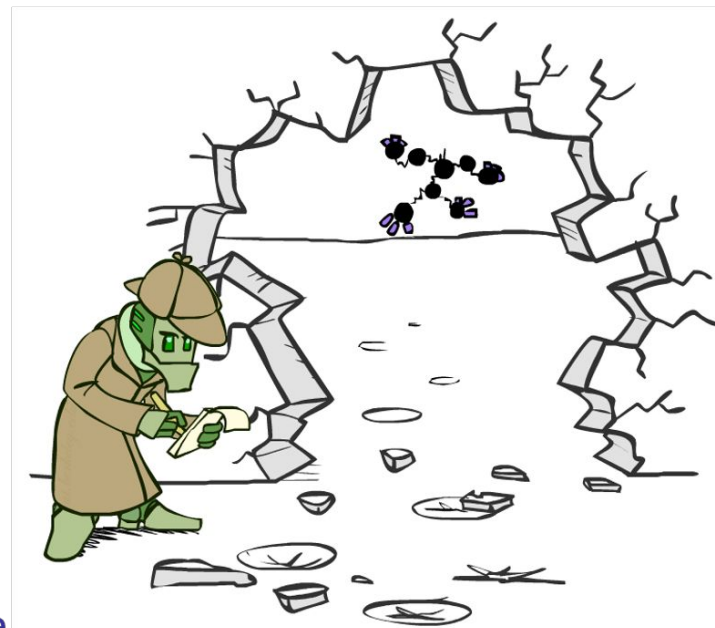
+r	0.1
----	-----

$$P(T|+r)$$

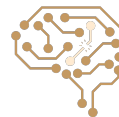
+r	+t	0.8
+r	-t	0.2

$$P(L|T)$$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9



- We eliminate all vars other than query + evidence



Value Elimination: Procedural Outline (cont.)

- Result will be a selected joint of query and evidence
 - E.g. for $P(L \mid +r)$, we would end up with:

$$P(+r, L)$$

+r	+l	0.026
+r	-l	0.074

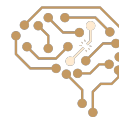
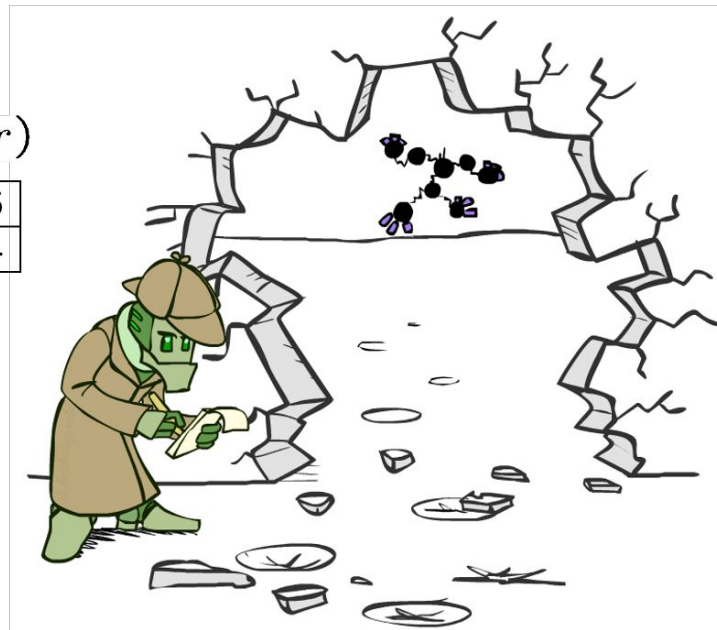
Normalize



$$P(L \mid +r)$$

+l	0.26
-l	0.74

- To get our answer, just normalize this!
- That's it!

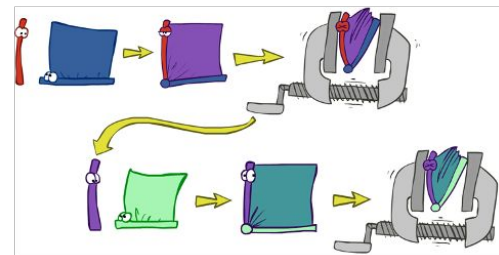


General Variable Elimination

- Query: $P(Q|E_1 = e_1, \dots, E_k = e_k)$
- Start with initial factors:
 - Local CPTs (but instantiated by evidence)
- While there are still hidden variables (not Q or evidence):
 - Pick a hidden variable H
 - Join all factors mentioning H
 - Eliminate (sum out) H
- Join all remaining factors and normalize



x	P(x)
-3	0.05
-1	0.25
0	0.07
1	0.2
5	0.01

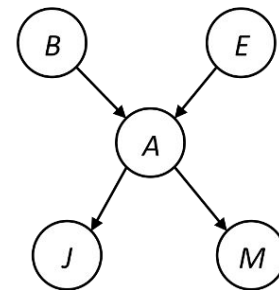



$$\times \frac{1}{Z}$$


Example

$$P(B|j, m) \propto P(B, j, m)$$

$P(B)$	$P(E)$	$P(A B, E)$	$P(j A)$	$P(m A)$
--------	--------	-------------	----------	----------

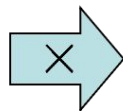


Choose A

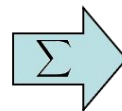
$$P(A|B, E)$$

$$P(j|A)$$

$$P(m|A)$$

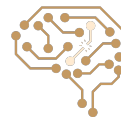


$$P(j, m, A|B, E)$$



$$P(j, m|B, E)$$

$P(B)$	$P(E)$	$P(j, m B, E)$
--------	--------	----------------



Example (cont.)

$P(B)$	$P(E)$	$P(j, m B, E)$
--------	--------	----------------

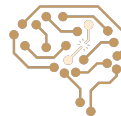
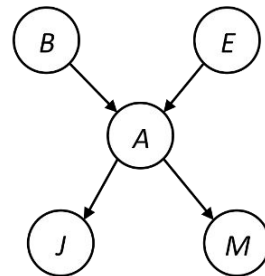
Choose E

$$\begin{array}{c} P(E) \\ P(j, m|B, E) \end{array} \xrightarrow{\times} P(j, m, E|B) \xrightarrow{\Sigma} P(j, m|B)$$

$P(B)$	$P(j, m B)$
--------	-------------

Finish with B

$$\begin{array}{c} P(B) \\ P(j, m|B) \end{array} \xrightarrow{\times} P(j, m, B) \xrightarrow{\text{Normalize}} P(B|j, m)$$

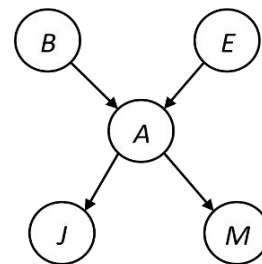


Same Example in Equations

$$P(B|j, m) \propto P(B, j, m)$$

$P(B)$	$P(E)$	$P(A B, E)$	$P(j A)$	$P(m A)$
--------	--------	-------------	----------	----------

$$\begin{aligned}
 P(B|j, m) &\propto P(B, j, m) \\
 &= \sum_{e, a} P(B, j, m, e, a) \\
 &= \sum_{e, a} P(B)P(e)P(a|B, e)P(j|a)P(m|a) \\
 &= \sum_e P(B)P(e) \sum_a P(a|B, e)P(j|a)P(m|a) \\
 &= \sum_e P(B)P(e)f_1(B, e, j, m) \\
 &= P(B) \sum_e P(e)f_1(B, e, j, m) \\
 &= P(B)f_2(B, j, m)
 \end{aligned}$$



marginal can be obtained from joint by summing out

use Bayes' net joint distribution expression

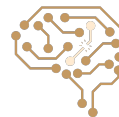
use $x*(y+z) = xy + xz$

joining on a, and then summing out gives f_1

use $x*(y+z) = xy + xz$

joining on e, and then summing out gives f_2

All we are doing is exploiting $uwy + uwz + uxy + uxz + vwy + vwz + vxy + vxz = (u+v)(w+x)(y+z)$ to improve computational efficiency!



Next Session: Sampling & HMM

